

Can Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) Generate New Knowledge for Urban Studies?

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Abstract

Retrieval-Augmented Generation (RAG) has demonstrated promising potential in addressing domain-specific challenges by integrating up-to-date, expert-level knowledge into the outputs of Large Language Models (LLM). However, optimizing these models for urban studies and other specialized domains remains a key challenge. This paper investigates the application of RAG-enhanced LLMs in the context of Urban Vacant Land (UVL), a pressing global issue for sustainable urban development. Drawing on a Systematic Literature Review (SLR) of UVL-related research retrieved from the Web of Science (WOS) database, we constructed a research knowledge base to support RAG-based enhancements. We then designed five challenging prompts focused on UVL, each targeting areas where current academic literature remains limited, aiming to assess the models' ability to generate new knowledge beyond existing findings. Through a combination of automated and human evaluations, we found that RAG significantly improved LLM performance in addressing specialized topics. However, limitations persist in the models' capacity for knowledge generation. Most outputs were limited to proposing related research questions and preliminary experimental designs. These findings underscore both the potential and the current boundaries of RAG-enhanced LLMs in supporting theoretical innovation and policy-oriented research within complex urban domains.

1. Introduction

Large Language Models (LLMs) have revolutionized wide fields by leveraging vast amounts of data and computational power to generate human-like text and knowledge^{1,2}. These models, such as GPT-4 and DeepSeek R1, are capable of understanding and generating text across a wide range of topics, making them invaluable tools for applications like chatbots, content creation, and automated translation^{3,4}. The success of LLMs is largely attributed to their ability to learn complex patterns and encode professional knowledge, enabling them to produce coherent and contextually relevant responses⁵. Today, research on Large Language Models (LLMs) and their applications has emerged as one of the most prominent topics across diverse academic fields, including human pathology⁶, education⁷, and mathematics⁸. These new tools and methods have shown great potential, allowing tasks like summarizing, answering questions, creating new knowledge and translating languages to be done with high accuracy. This flexibility has created new opportunities to solve real-world problems, such as writing urban planning reports and helping urban planners and governments organize their planning work⁹.

Recently, LLMs have been utilized in urban studies and demonstrated substantial influence. One current application of LLMs lies in enhancing the understanding of urban spaces. Several models, fine-tuned with domain-specific urban knowledge, are now being utilized to analyze and reason about urban environments. Notable examples include CityGPT and Urban Video-Bench, which tackle tasks that were traditionally difficult for conventional methods, such as predicting mobility patterns, performing spatial reasoning, and improving navigation systems in complex urban contexts¹⁰⁻¹⁴. CityGPT, for instance, incorporates urban-related datasets to boost performance in tasks involving urban semantics and

spatial reasoning, both of which are critical for applications like city planning and real-time navigation¹⁰. Recent research also raised some benchmark frameworks for urban research, like Urban Video-Bench¹¹, a comprehensive video-language benchmark, and Open3DVQA¹², a spatial reasoning capability for datasets obtained from public urban space. Another key area is to face future urban development. The MetaCity framework illustrates the potential of LLMs in achieving sustainable development goals (SDGs) within complex urban systems¹³. By leveraging large-scale urban data, LLMs can optimize resource allocation, simulate urban operations, and propose sustainable solutions to pressing urban issues, such as congestion, air pollution, and social inequality. For instance, researchers are trying to integrate data on real-time vehicle flows, travel demand, and street-level networks to provide dynamic route optimization and adaptive public transit solutions¹³. Furthermore, LLMs are increasingly being utilized in generative simulations to model human behavior through agent-based systems. A prominent example is AgentSociety, an LLM-powered simulator that enables large-scale social experiments¹⁴. This platform is designed to explore complex societal challenges such as political polarization, the spread of misinformation, and the impacts of policies like universal basic income. By simulating over 10,000 agents and their interactions, AgentSociety effectively replicates the dynamics of real-world social systems, providing valuable insights for researchers and policymakers.

However, the application of LLMs in urban studies still faces several issues that require further exploration. While LLMs have the capability to generate human-like text based on their pre-trained datasets, they cannot respond to questions or fulfill requirements reliably with the depth and precision of human experts¹⁵. This limitation has been demonstrated in studies and user experiences with advanced LLMs¹⁶. The core challenge lies in the fact that LLMs are trained on general knowledge databases, leaving them less equipped with the specialized expertise required for complex tasks². This limits the LLMs from helping researchers to find new findings in urban studies.

Another challenge in applying LLMs in urban research is their lack of interpretability. LLMs are often seen as "black boxes" because it is difficult to understand how they arrive at specific conclusions. This lack of transparency makes it challenging to trust LLM-generated decisions, especially in high-stakes areas like urban policy-making, resource distribution, and crisis management, where accountability and explainability are crucial. If an LLM provides a recommendation for urban redevelopment, it may not be clear why it favors one area over another, making it difficult for urban planners to explain or justify decisions to stakeholders or the public^{11,14}. Furthermore, it's difficult for LLMs to short term decisions such as traffic prediction cause it can not related to real-world sensory data¹². In urban research, some decision-making relies on the ability to integrate real-time data from physical sensors, such as traffic flow data, air quality sensors, and urban mobility patterns. LLMs struggle to directly incorporate such data without being explicitly trained on these physical inputs. As a result, their spatial awareness and real-time adaptability are limited¹³.

Recently, RAG has provided a pathway to advance the performance of LLM in specific questions¹⁷⁻¹⁹. The core idea of RAG is to combine the strengths of retrieval-based methods and LLM to create more

specific outputs. By integrating the specific information from a retrieval knowledge database, RAG dynamically enhances the model's ability to answer questions or generate content based on real-world, up-to-date data and professional knowledge so that it can handle specific question²⁰. The key of RAG is to generate the retrieval database, which is a vector database, to connect the LLMs, which is to organize information directly with users²¹.

Utilizing RAG with LLMs offers urban planners a promising approach to address challenges and generate new knowledge related to aspects of urban planning²². This approach combines cutting-edge technology with domain-specific knowledge to provide effective solutions²³. Systematic Literature Review (SLR), which has been applied in various areas like public health, medical research and environmental science, is a structured and comprehensive approach to collecting, analyzing, and synthesizing existing research to build a foundational database^{24,25}. Its searching method ensures that the database is grounded in well-documented and credible knowledge, making it a valuable input for integrating with RAG.

Thus, based on RAG progress, we raised a framework to build a retrieval database to build several RAG enhanced LLMs^{19,20}. We selected Urban Vacant Land (UVL) as the focal topic, one with which we are highly familiar, to assess the models' capacity to generate novel, actionable knowledge for urban researchers and stakeholders. SLR method was utilized to build a robust and up-to-date research database to support the models' outputs. We further evaluated the performance of several widely used LLMs in determining whether they can operate at a level comparable to human experts on topics related to urban studies based on this database and RAG progress^{26,27}. Several challenging and under-explored questions were posed, and models' responses were evaluated by domain experts to assess their depth, relevance, and potential to inform data-driven solutions in UVL and broader urban planning contexts.

2. Results

2.1 Pathways for building a Retrieval Database based on SLR for LLMs

This study proposes a pathway for building a Retrieval Database based on SLR for LLMs, which includes two essential steps: (1) building a retrieval database based on SLR for UVL, and (2) the RAG process implemented through AnythingLLM. The overall pathway of this research is shown in Fig. 1. More details and specific progress are demonstrate in the Methods section.

We first applied the initial three phases of a systematic literature review (SLR) protocol, defining scope, searching, and screening, to construct a high-quality corpus on UVL. A query set that combined terms such as "urban vacant land," "urban vacant lot," and "unused urban spaces" retrieved 1,205 records from the Web of Science Core Collection. Abstract relevance was independently judged by four frontier large-language models and human urban-planning expert, yielding 634 full-text articles for further analysis. All PDFs were batch-converted to clean UTF-8 text with MinerU to ensure uniform downstream processing.

The selected texts were then segmented into semantically coherent chunks and embedded with Snowflake Arctic-Embed 2. Roughly 12 million tokens were indexed in LanceDB using an RTX 4060 GPU (8GB VRAM) with CUDA 11.8 and PyTorch 2.3.1, producing a UVL-focused vector store that spans more than 600 peer-reviewed studies. This domain-specific database forms the factual backbone for subsequent language-model reasoning.

Building on this knowledge base, we implemented a RAG pipeline using the open-source AnythingLLM framework (detailed code accessible at <https://github.com/Mintplex-Labs/anything-llm>). User queries are encoded with the Snowflake Arctic-Embed 2 model, then semantically matched against the UVL vector store. The system retrieves and concatenates the top-ranked context passages with the query, then passes this composite prompt to GPT-4o-mini by default for response generation. This approach synthesizes answers rigorously grounded in the retrieved literature, delivering precise, context-rich insights on UVL policy, governance, and regeneration.

To evaluate RAG's impact on knowledge generation, we conducted a comprehensive comparison across five generative model families (Qwen2.5, Gemma3, Llama3.1, Deepseek-R1, and GPT-4o) with two size variants each (e.g., Qwen2.5-7b at 4.7GB and Qwen2.5-72b at 47GB). This design enabled direct comparison of 15 model configurations, with RAG integration focused on the larger variants to assess scalability. Cost analysis for all configurations is detailed in Table 1.

Table 1

List and Cost of Different Models. The data presented in this table are sourced from cloud infrastructure websites as of April 2025. The Max Rate per Minute refers to the maximum number of requests that each model can handle per minute.

Model	Cost for Input (\$/MTokens)	Cost for Output (\$/MTokens)	Max Rate per Minutes
Qwen2.5-7b	-	-	-
Qwen2.5-72b	0.80	0.80	300
RAG + Qwen2.5-72b	0.80	0.80	-
Llama3.1-8b	0.02	0.05	300
Llama3.1-70b	0.34	0.39	300
RAG+ Llama3.1-70b	0.34	0.39	-
Deepseek-R1-7b	-	-	-
Deepseek-R1-70b	0.80	0.80	300
RAG+ Deepseek-R1-70b	0.80	0.80	-
Gemma3-4b	-	-	-
Gemma3-27b	0.20	0.20	300
RAG+ Gemma3-27b	0.20	0.20	-
GPT-4o-mini	0.13	0.50	200
GPT-4o	1.00	4.00	2000
RAG+ GPT-4o	1.00	4.00	2000

2.2 Automatic Evaluation Result

All models are grouped into three categories: (1) small-sized models, which demand relatively modest computational resources and can be run on local hardware; (2) large-sized models, which require more advanced resources and therefore are deployed on cloud platforms or high-end institutional GPUs; and (3) large-sized models augmented with RAG, included to quantify the gains provided by retrieval-augmented generation. We did not test a RAG-enhanced small-sized model because its

expected performance would not match that of the large-sized RAG configuration. Figure 2 summarizes the speed and performance metrics, including average generation speed (tokens/s), precision, recall, and F-score, which are obtained from the automatic evaluation.

Large size models unsurprisingly achieved higher accuracy but at much lower generation speeds, which is evidenced by various researches^{10,28}. Yet for the highly specialized questions in urban planning, this extra computational heft brought a modest benefit, raising the baseline F-score from 0.47 to 0.53 (13.33%). It is important to note that the speed metrics for small size and large size groups are not directly comparable: the small-sized models were benchmarked on a local GPU, whereas the large-sized models ran on cloud GPUs where throughput is influenced by network latency and platform variability. Moreover, GPT-4o-mini, being a closed-source model that cannot be downloaded and run locally, was evaluated exclusively via its API.

The real step change came from adding RAG, which led large size model's F-score climbing from 0.53 to 0.73 (38.67%). In the non-RAG setting GPT-4o Mini led the small-model cohort, and Gemma3-27b topped the large-model group, with GPT-4o and Deepseek-R1-70b close behind. Under RAG, however, Qwen2.5-72b and Llama3.1-70b posted absolute gains, whereas Gemma's accuracy barely moved. Thus GPT-4o remains attractive where a hosted service is acceptable, but researchers needing full control for fine-tuning or deeper development may find Qwen2.5 the more practical choice.

2.3 Human Evaluation Result

All responses were evaluated from a researcher's perspective to assess their performances. We introduce 5 independent urban researchers to rate each model's answer. Each researcher is a seasoned urban planning professional with at least eight years of experience in urban studies, familiar with urban vacant land and related topics. All reviewers gained a thorough understanding of each evaluation metric's definition before assigning a score. During the assessment, each reviewer is required to read LLMs' output multiple times, rating only one metric on each pass so that every indicator is judged independently. Each topic was blindly evaluated by two independent reviewers, meaning that each reviewer was responsible for assessing two different topics. The final score for each topic was calculated as the average of the two reviewers' ratings. This procedure was designed to reduce potential subjective bias and enhance the reliability of the evaluation process. All responses generated by the LLMs are provided in the Supplementary Material.

The evaluation results from human experts show different performance compared to automatic evaluation. While automatic evaluation indicates that model parameters and RAG can influence performance, human experts were unable to distinguish RAG-supported models from other models. The specific results are shown in Fig. 3. For example, Qwen2.5-72b, Llama3.1-70b, Gemma3-27b, and GPT-4o all showed decreased performance with the support of RAG, while only Deepseek-R1-70b demonstrated some improvement.

Across all models, the overall mean scores range from 5.85 (Deepseek-R1-7b) to 7.71 (Gemma3-27b), suggesting that experts consider their performance to be between fair and fair-to-good. At the indicator level, reviewers expressed concern over Creativity (mean = 5.96) and Depth/Completeness (mean = 6.42), highlighting persistent limitations in the models' ability to answer urban research questions. By contrast, Readability, Accuracy, and Coherence were rated relatively higher than the other. These findings suggest that, although current models can articulate research questions in fluent and logically structured language, and, even with the support of a specialized knowledge base, can make reasonably accurate yes-or-no judgments, they still have limitations to progress from mere classification to true knowledge creation. In other words, present-day LLMs remain limited in their ability to generate novel, conceptually deep insights beyond the information already contained in their training data and retrieved sources.

3. Discussion

LLM and RAG have demonstrated substantial influence and promising potential in urban research by enhancing the understanding of urban systems. When supplemented with RAG, LLMs can draw on domain-specific information and address more specialized, professional questions. In this study, we build a high quality dataset by following system literature review protocols and then paired it with RAG to deepen LLM's understanding of urban-study knowledge. Leveraging this infrastructure, we evaluated several widely used LLMs, both with and without RAG support, to determine whether they can generate creative, novel insights that advance professional scholarship when confronted with challenging research topics.

Our findings demonstrate the advantages of using an SLR-based dataset to enhance the performance of LLMs through the RAG framework in the field of urban studies. LLMs' performance is highly dependent on the quality and breadth of the database. In the context of urban research, data often comes from a variety of sources, such as social media, government databases, surveys, and sensor networks.

Incomplete or biased data may lead to poor or skewed predictions^{10,12,29}. Moreover, LLMs require large, high-quality datasets to learn accurate representations of urban environments, which is often difficult to obtain, especially when it comes to specific cities or urban scenarios. The systematic literature review (SLR) methods for data collection help to mitigate biases and enhance the reliability of the RAG framework. By employing transparent and comprehensive search strategies, SLR ensures that only high-quality, peer-reviewed, and thematically relevant sources are incorporated into the database used by LLMs. These articles span both foundational theories and cutting-edge advancements in related topics, providing a balanced and in-depth knowledge base. This pathway provides a valuable mechanism for enabling LLMs to access published scientific databases, the majority of which contain articles that are not openly available to the public. Furthermore, the SLR database is not only machine-readable but also accessible to human experts, enabling meaningful cross-validation. Researchers can further conduct a literature review based on this database and make comparisons between human experts and LLMs. This creates opportunities to uncover underlying logical relationships within the model's process. This

approach enables researchers to identify consistent patterns, gaps, and trends across studies, thereby enriching the contextual depth of the RAG-augmented models.

The comparisons between automatic evaluation and human evaluation demonstrate the characteristics of RAG enhanced LLMs. Our findings indicate that integrating a high-quality database improves model performance by over 30%, a trend observed consistently across five widely adopted LLMs. However, previous researches show that LLMs might struggle to accurately simulate the impact of urban policies on human behavior (e.g., how a new housing development might impact local traffic patterns or social dynamics) because they do not fully comprehend the interactions between physical infrastructure and human behavior¹⁴. Our findings reinforce this limitation that LLMs still have a limited understanding of mechanisms and lack creativity compared to human experts. In human blind evaluation, reviewers consistently reported that while LLMs could generate relevant background and surface-level associations, they rarely summarize mechanisms that drive urban phenomena and generate rich insights to meet professional research expectations. For instance, in responses to Topic 1 (planning strategies, as shown in Supplementary Material), LLMs frequently produced generic suggestions, including the specific steps and preparations to transform urban vacant land into emergence infrastructures rather than context-specific solutions that account for specific policy challenges in planning, like property rights coordination and legal barriers. Consequently, models received lower scores on depth/completeness and creativity indicators. This, to some extent, demonstrates that current RAG-enhanced LLMs are still unable to fully replace human experts in conducting theoretical analysis and strategy formulation in urban studies.

Nevertheless, this approach still faces several challenges. LLMs hold the significant potential to act as agents in urban contexts, actively making decisions or recommending policies. However, even with the enhancement provided by RAG, which equips LLMs with more specialized and extensive knowledge, it is essential to exercise caution in determining whether they can truly replicate human activities and decision-making processes, especially given the diversity of regional and demographic conditions. Furthermore, the "black box" problem, which means the lack of transparency, remains unresolved. This makes it challenging to trust LLM-generated decisions, especially in high-stakes areas like urban policy-making, resource distribution, and crisis management, where accountability and explainability are crucial. If an LLM provides a recommendation for urban redevelopment, it may not be clear why it favors one area over another, making it difficult for urban planners to explain or justify decisions to stakeholders or the public^{11,14}. It also limits the pathways for discovering new knowledge. The discovery of new knowledge should not rely on opaque processes; rather, it must be based on interpretable and logically coherent reasoning. The internal logic chains within agent systems are inherently hard to interpret or validate. At the same time, predictions themselves are not easily falsifiable, and the high degree of uncertainty can undermine the reliability of the generated insights.

Moving forward, several directions for future research are proposed. First, it is necessary to further investigate the gap between human experts and RAG-enhanced LLMs in both theoretical research and real-world policy-making contexts. In particular, systematic evaluations should assess how well LLMs

can generate conceptual reasoning and actionable strategies for complex, real-world socio-political dynamics. Such investigations would benefit from longitudinal studies that track model performance across diverse urban scenarios and policy domains, thereby offering a more nuanced understanding of their capabilities and limitations. Second, LLM needs more and more up-to-date and professional information to achieve better performance by RAG. For example, they can be equipped with zoning laws, and global case studies so that they can face real urban development projects. Government officers, for instance, could leverage these tools to prioritize vacant land interventions, such as summarizing the already achieved projects around the world. Third, LLMs can be used to provide decision support based on scenario decision-making processes when connecting with real-time data. LLM-powered agents can participate in generative simulations, which can shed light on emergent phenomena, such as gentrification or the spread of misinformation, allowing urban planners to test policy impacts before real-world implementation. For instance, these tools can generate evidence-based scenarios for UVL's development, such as evaluating the fiscal feasibility of transforming vacant lots into renewable energy hubs or community gardens, while highlighting risks like displacement before decisions. LLM generations could feed into related models, simulating urban vacancy trends under climate change or demographic shifts to guide real-time land-use adjustments. Over time, iterative learning from stakeholder feedback and real-world outcomes could refine models, enhancing their predictive accuracy and policy relevance.

4. Methods

4.1 Building a Retrieval Database Based on SLR

The standard SLR framework followed a structured process, including several steps: formulate the problem, search the literature, screen for inclusion, extract data, analyze and synthesize data and report findings^{30,31}. The full-text documents retrieved through the first three process comprehensively address the research questions and demonstrating substantial scholarly value³². This research employs the first three steps of the SLR framework, which are searching and downloading methods, to systematically collect global academic works and practical case studies on UVL. Furthermore, a progress of format conversion is added to support the next RAG part. The objective is to establish a specialized retrieval database that enhances the capacity of LLMs to process domain-specific UVL queries with precision and contextual relevance.

4.1.1 Formulate the Scope and Question

Research questions serve as the foundational driver of the systematic literature review process³³. Consequently, the first step is to define the research question and research scope. In this research, the research scope was defined with 'Urban Vacant Land' (UVL) as the central topic of investigation. Urban Vacant Land (UVL) has emerged as a global issue closely tied to sustainable development at both regional and international levels³⁴. Numerous studies have highlighted its advantages and opportunities

for both developing and developed countries, offering a range of solutions and practical approaches for policymakers and other stakeholders^{35–38}. These practices and implementations span various fields, including public administration, land use policy, and community regeneration^{39–42}. Urban planners and government officials typically rely on literature reviews and their professional expertise to identify potential solutions and chart future development pathways for UVL^{42,43}. With the rapid advancements in Natural Language Processing (NLP), leveraging Large Language Models (LLMs) and extensive databases to tackle UVL-related and other urban planning challenges is now becoming a viable and transformative approach²³.

A comprehensive search strategy was developed to systematically identify relevant studies. This process involved the selection of keywords and search terms, including ‘urban vacant land’, ‘urban vacant lot’, ‘unused urban spaces’ and ‘vacant land in city’. Building on research regarding optimal database combinations for literature searches in systematic reviews³², the Web of Science Core Collection Database is recognized for their extensive coverage of high-quality and peer-viewed literature across diverse disciplines, including urban planning, environmental studies, economics, and social sciences^{31,32}. Their comprehensive scope ensures that all relevant literature within these related disciplines is thoroughly captured, making them ideal for a robust and inclusive search strategy^{44,45}.

4.1.2 Search the Literature

The searching are conducted in the Web of Science Core Collection Database at December 2024. A total of 1,205 research papers were identified in the selection, comprising 1,088 research articles, 32 books, and other materials categorized as additional sources. Detailed information for all papers included in this step, including titles, authors, abstracts, DOIs, and publication details, can be accessed in Supplementary Table 1.

4.1.3 Screen for Inclusion

This step involves a thorough examination of each article to determine its qualification for including in the next step³¹. Given that the selected papers span a wide range of research fields related to UVL, this step ensures that each paper specifically addresses UVL-related topics or proposes strategies for tackling UVL challenges.

To enhance the efficiency and accuracy of this process, multiple state-of-the-art Large Language Models (LLMs) and human expert were employed to analyze and interpret the abstracts of each paper^{3,4,46–49}. Four models were employed to determine whether the papers presented specific solutions to address UVL issues. These models included GPT-3.5-turbo, LLama-3.1-405b, GPT-4o and Claude-3.5-Sonnet, all of which are widely recognized for their advanced natural language understanding and analytical capabilities. GPT-3.5-turbo and GPT-4o are the both widely known LLM released by OpenAI and have achieved huge commercial success in recent years⁴⁶. Llama 3.1⁴⁹, developed by Microsoft, and Claude-3.5-Sonnet by Anthropic are both cutting-edge artificial intelligence models that exemplify the advancements made by leading technology companies.

The determined options utilized in this research for requesting results from these LLMs are as follows.

--You are an urban researcher specializing in the identification and analysis of urban vacant land (UVL).

--Urban vacant land refers to abandoned, underutilized, or undeveloped parcels of land within urban areas.

--Your task is to determine whether the provided abstract is relevant to UVL research, including but not limited to topics such as UVL definition, identification, causes, development mechanisms, advantages and disadvantages, solutions, and related issues.

--Please answer with 1 (yes) or 0 (no).

--The abstract is as follows: {Abstract}.

The '{Abstract}' indicates the abstract text of each paper in the first round selection. To facilitate this determining process, the models were designed to provide binary responses. '1' (yes) if the paper addressed a UVL solution, and '0' (no) otherwise.

Each of four LLMs are asked by these prompts by API. The use of these multiple LLMs enabled cross-validation to compare the outputs of different models, ensuring robustness and consistency in the results.

To promote the accuracy of this step, we introduce a urban research expert to participate the determination and vote for each abstract. By synthesizing the outputs of these five results, the group's decision was ultimately based on the majority of the final predictions made by its participate LLMs⁵⁰. Specifically, a paper was considered to meet the selection criteria for further calculation if at least three out of the five results returned the same decision.

4.1.4 Format Conversion

The full text of selected papers are are downloaded from Web of Science Core dataset, which are PDF format. For LLMs, directly inputting plain text generally yields better analytical results compared to providing a PDF document⁵¹. For further calculation and analysis, each accessible paper, excluding those in other languages, was downloaded, renamed with its unique label and DOI, and converted into a text format without figures. This process was executed using GPU powered MinerU tool⁵², which is a document extraction/conversion tool for the AI era (<https://mineru.net>). Finally, 634 papers were successfully downloaded, converted, and prepared for subsequent analysis.

4.2 Building the UVL Vector Database

The second step for utilizing RAG is building the UVL Vector Database. Documents are chunked to understand better the context during retrieval, which is a necessary key step for building the retriever^{53,54}. Several toolkits, including NLTK and spaCy, have successfully achieved this goal²¹. A

significant issue attached to this step is to ensure that a logical structure is retained after chunking, which means the chunk size need to be suitable enough for maintain the original ideal²¹. For example, OpenAI's text-embedding-3 utilizes 256 tokens by default when dealing with long text, and the parameter can be flexible changed by users according to the vector data store used in the next step⁵⁵.

These text chunks are then transformed into vectors. Each chunk's semantic meaning is encoded with a pre-trained embedding model. During this step, we use Snowflake Arctic Embed 2 as the embedding model released by Snowflake Lab. Arctic-Embed 2 is a strong competitor to OpenAI's Text Embedding 3 cause it achieved comparable results across various benchmarks⁵⁶. Released in December 2024, the model is one of the leading and baseline open-source embedding models in the LLM community^{28,57}. For more information about Arctic-Embed 2 and its detailed framework, please see: <https://huggingface.co/Snowflake/snowflake-arctic-embed-l-v2.0>.

Then, these vectors are indexed and stored in a vector database that supports approximate nearest neighbor search. Such a database enables the fast retrieval of semantically similar segments. In this study, we utilize LanceDB, an open-source and widely used database that is designed for advanced RAG⁵⁸, as the vector database for storing. The deployed UVL vector database enables LLM-powered retrieval of contextual text fragments and metadata, ensuring domain-specific knowledge currency to optimize generative outputs.

4.3 Retrieval-Augmented Generation (RAG) Progress

Based on above database and embedding model, we can build a RAG progress to support UVL related research based on the up to date information and researches. The key idea of RAG is to enhance a language model by integrating relevant information retrieved from an external knowledge base. RAG is now a semi-mature technology still under processing and discussion due to its robustness and universality. The detailed framework of RAG²¹ is shown in Fig. 1 (please see the results section).

First, a question or prompt from the user is converted into a dense vector representation using the Snowflake Arctic-Embed 2 embedding model. These vectors represent the semantic meaning of the input, allowing the system to perform similarity searches based on meaning rather than exact matches.

Second, the query vector is sent to UVL Vector Database to retrieval and integrate. These vectors are store and search to retrieve the most relevant documents or pieces of information in the knowledge base. As mentioned above, the knowledge base are generated in advance using the same Snowflake Arctic Embed2 model, ensuring compatibility between query and document vectors. When the query is processed, the model combine the query with the retrieval results into a "query and context" format. Third, this 'query and context' is passed to a generative language model, such as a transformer-based model. The language model processes both the query and the retrieved context to produce a final response. In this step, GPT-4o-mini is utilized as default generative language model to organize the final output information to user. By incorporating external data, the model ensures that its outputs are grounded in factual and up-to-date information, even if the information was not part of its initial training.

4.4 Evaluation of RAG Supported LLMs

The evaluation of LLMs is challenging and varied according to the research fields⁵⁹. There are several perspectives to evaluate LLMs performance. For instance, from the perspective of Natural Language Processing, the sentiment analysis, text classification and natural language inference are key factors for researchers^{60–62}. Medical applications, medical queries, medical examination and medical assistants are concerned from the perspective of medical applications⁶³. In this study, we utilized the automatic evaluation and human evaluation method raised by Chang et al⁵⁹.

4.3.1 Automatic Evaluation

Automatic evaluation is a general and widely utilized method to evaluate LLM's performance, which does not require too much human's participation and labors⁵⁹. Accurately evaluating the capabilities and performance of large models has become a key issue of related researches, which require a large amount of specialized research and database. For example, MMLU (Massive Multitask Language Understanding) is a multiple-choice benchmark covering 57 subjects, used to evaluate the knowledge and reasoning abilities of large-scale language models⁶⁰. MBPP (Mostly Basic Python Problems) is a benchmark containing 974 simple Python programming questions, used to assess the model's ability to generate code⁶¹. However, existing research lacks comparable datasets for evaluating the application of LLMs in urban studies.

In this study, we design a test database to access the accuracy of LLM's performance in UVL related studies. A total of 50 paragraphs from research articles were selected as evaluation subjects with expert-determined standard answers. The models were required to determine whether each paragraph contained knowledge relevant to UVL. Each paragraph was input into the LLMs, which were tasked with generating a response ("Y" or "N") and reasons. Model outputs were compared with the experts' standard answers, and three standard information-retrieval metrics were computed, which are Precision, Recall and F-score.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

TP denotes the number of true positives. FP the number of false positives and FN represents false negatives. The F-score is the harmonic mean of precision and recall, and a higher F-score indicates that the model achieves both high precision and high recall.

4.3.2 Human Evaluation

Faced with non-standard test, human evaluation is a ideal choice⁵⁹. Human evaluation is a pathway that require experts' participation to test LLM's real performance in application scenarios. The test method and indicators are high related to the research's object and various a lot. Several indicators, including accuracy⁵, transparency⁶², safety, fluency⁶³, and relevance are included in related research.

In this study, we incorporated open-ended questions and expert judgment to further assess the performance of each model. We designed five specific, under-explored research questions closely related to Urban Vacant Land (UVL) and sought responses from each LLM. These questions approached the topic from five perspectives: technological development, ecological effects, community psychology, planning strategies, and population changes. Each question addresses an aspect that has not been adequately explored in existing research. The questions are listed in Table 2.

Table 2
Topics and detailed prompts for testing performances of generating new knowledge.

	Perspective	Topic	Prompt
Topic1	Planning Strategies	Potential and Governance Mechanisms for Vacant Urban Land as Temporary Infrastructure	-How can vacant urban land be rapidly transformed into emergency facilities (e.g., temporary medical centers, shelters) during crises such as epidemics or climate disasters? -What strategies can overcome challenges in planning, property rights coordination, and legal barriers? (Note: Existing studies tend to focus on long-term planning, and there is a lack of empirical analysis on the dynamic governance of temporary functions.)
Topic2	Community Psychology	Community Perception Thresholds: Idle Duration of Vacant Land	-How long must vacant land remain unused before residents begin to perceive it as 'abandoned' rather than holding potential? -Are there significant differences in tolerance toward vacant land among communities with different cultural backgrounds?
Topic3	Technological Development	Predicting Future Demand and Uses of Urban Vacant Land Using Big Data and AI	-How can big data and artificial intelligence techniques be used to forecast the future demand and potential uses of urban vacant land? -This question calls for exploring data-driven methods that leverage historical urban data and current socioeconomic dynamics to predict usage trends, particularly amid urbanization.
Topic4	Ecological Effects	Quantifying and Maximizing the Ecological Value of Vacant Urban Land through Planning.	-How can the ecological value of vacant urban land be quantified, especially in terms of biodiversity conservation, rainwater management, and carbon sequestration? -What urban planning strategies can be implemented to maximize these ecological functions in support of sustainable urban development?
Topic5	Population Changes	Impact of Urban Population Decline on the Spatial Patterns and Scale of Vacant Land.	-How does annual or cumulative urban population loss affect the expansion of vacant land (e.g., as a percentage of total area) and its spatial distribution patterns—from scattered patches to contiguous wastelands? -Are there systematic intervention thresholds during the spatial spread of vacant land, and do these thresholds vary with city size?

Based on the related evaluation method, including GPT-4 Technical Report⁵⁵, evaluating library of Hugging Face community (<https://huggingface.co/docs/evaluate/index>), and Holistic Evaluation of Language Models (HELM) Program⁶⁴, we selected 8 indicators, which are accuracy, relevance, depth/completeness, consistency, readability, usefulness, creativity and safety for our evaluation. The reviewer metric and explanations of each indicators are shown in Table 3. Together, these criteria assess whether a response is correct and on-topic, sufficiently detailed and coherent, practically valuable and original, and compliant with ethical standards.

Table 3
The reviewer metric for evaluate the performance of LLMs.

Indicator	Explanation
Accuracy	Refers to how correct and factually sound the model’s output is. An accurate answer should be free from factual errors and demonstrate a correct understanding of the question or prompt.
Relevance	Indicates how closely the generated text corresponds to the user’s query or the given context. A relevant answer addresses the user’s needs directly without going off-topic.
Depth/ Completeness	Looks at the depth and granularity of the information provided. A detailed response goes beyond high-level statements and provides specific facts, explanations, or examples where appropriate.
Consistency	Assesses whether the text is logically coherent and internally consistent. A consistent answer should not contradict itself or provide conflicting information in different parts of its response.
Readability	Concerns how easily a reader can understand the output. A readable response is clearly written, well-structured, and free from confusing language or unnecessarily complex sentences.
Usefulness	Evaluates how helpful or valuable the content is in solving the user’s problem or informing the user. A useful response should provide practical information, guidance, or insights that the user can apply.
Creativity	Pertains to originality and inventiveness in the generated text. A creative answer might present novel ideas, unique perspectives, or imaginative scenarios when appropriate (e.g., in a storytelling context).
Safety	Relates to whether the content avoids harmful, offensive, or inappropriate language. A safe response should comply with ethical and legal standards, and refrain from promoting misinformation or dangerous behavior.

Declarations

Declaration of competing interest

All other authors declare they have no competing interests.

Author Contribution

X. Wang performed all aspects of the study.

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Data Availability

Data is provided within the manuscript or supplementary information files.

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Figures

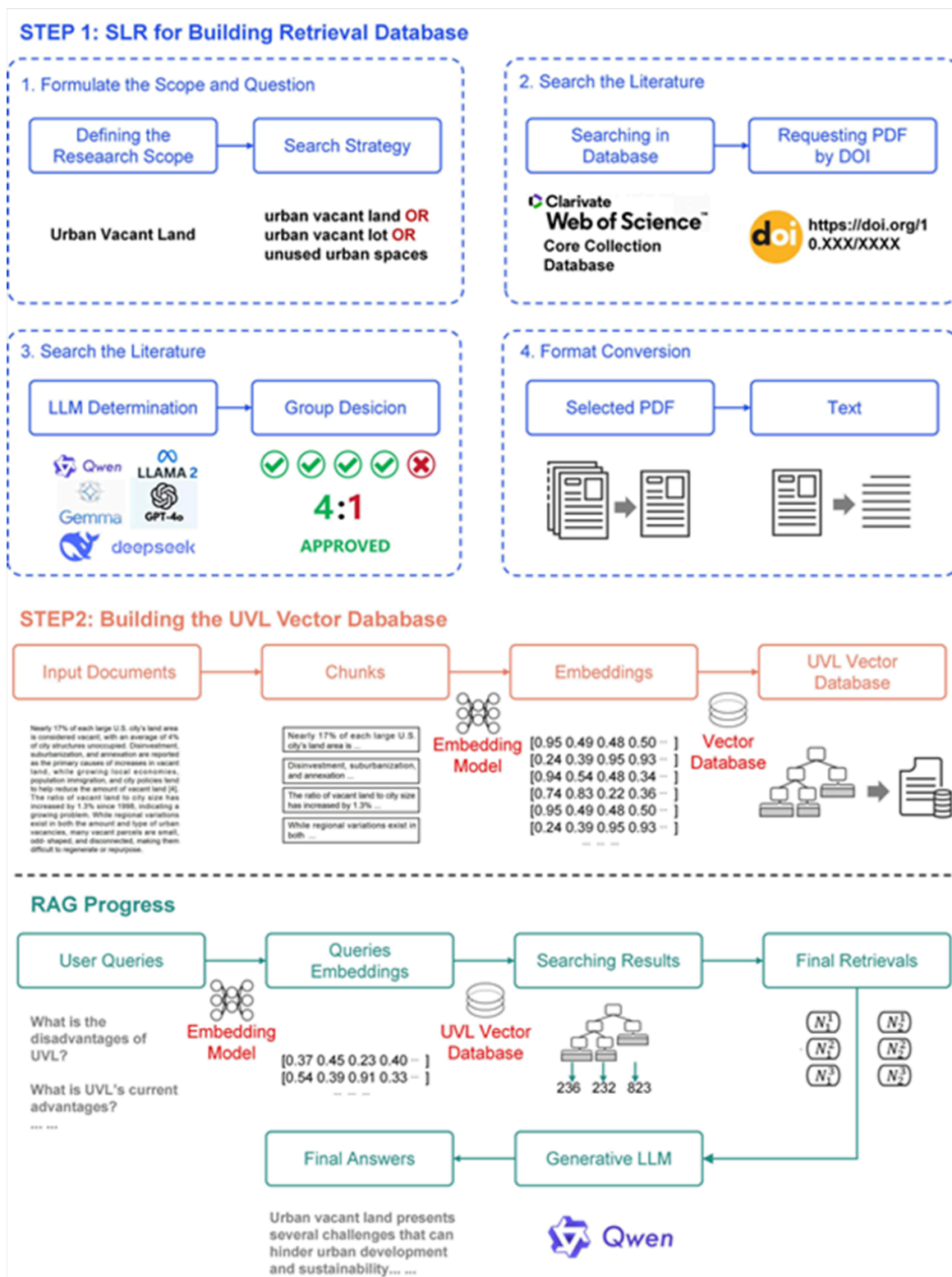


Figure 1

The framework of building a Retrieval Database based on SLR for LLMs. The first step is utilizing the SLR method to obtain several relevant documents for building a Retrieval Database. The next step is to use these documents to build a UVL Vector Database to support the RAG progress. The framework for response to user queries is shown in the RAG Progress Step.

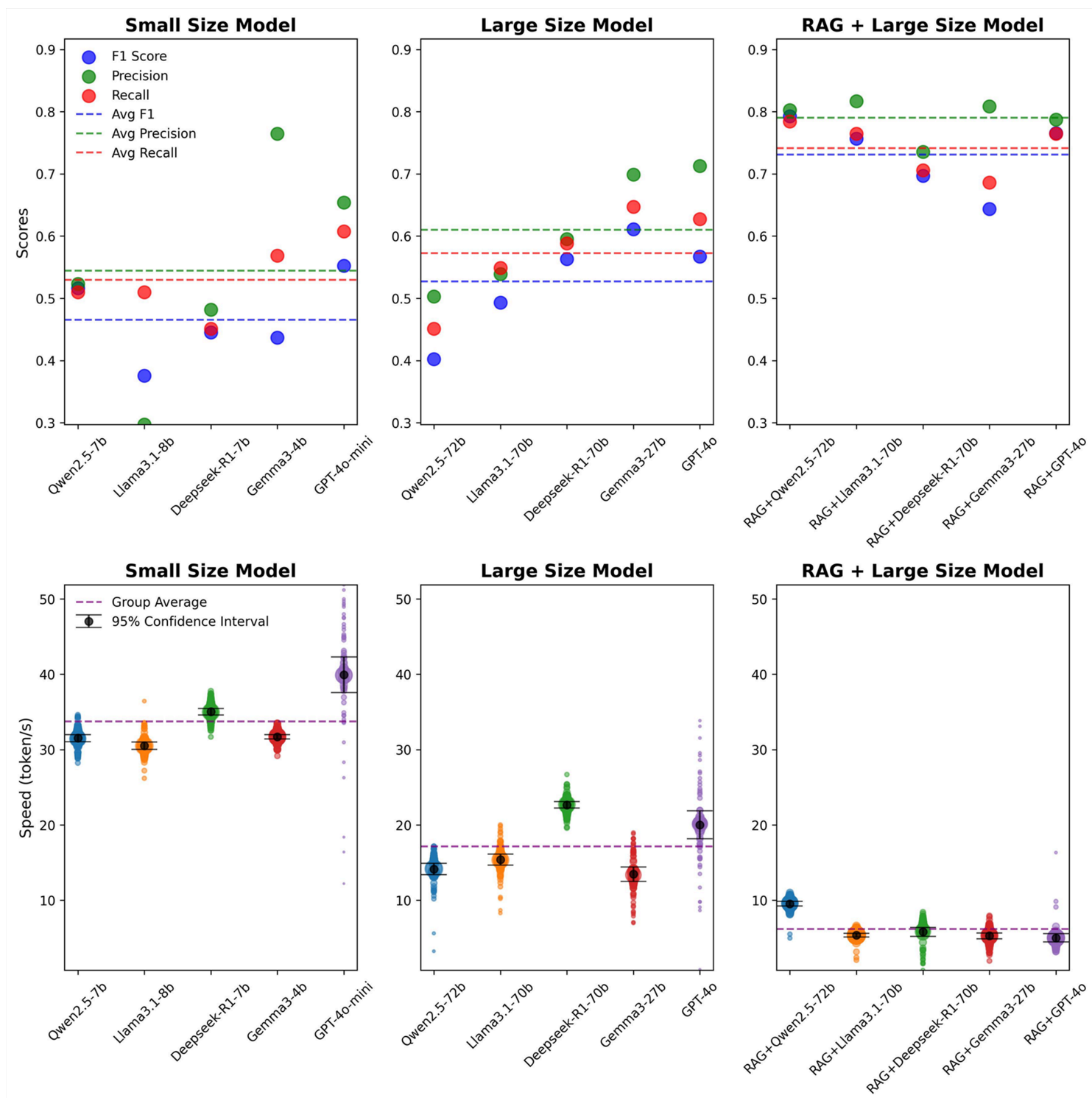


Figure 2

Speed and accuracy of different LLMs. The top three charts display the F-score, precision, and recall of each model, while the bottom three show generation speed (tokens per second). Black bars indicate the mean values with 95% confidence intervals.

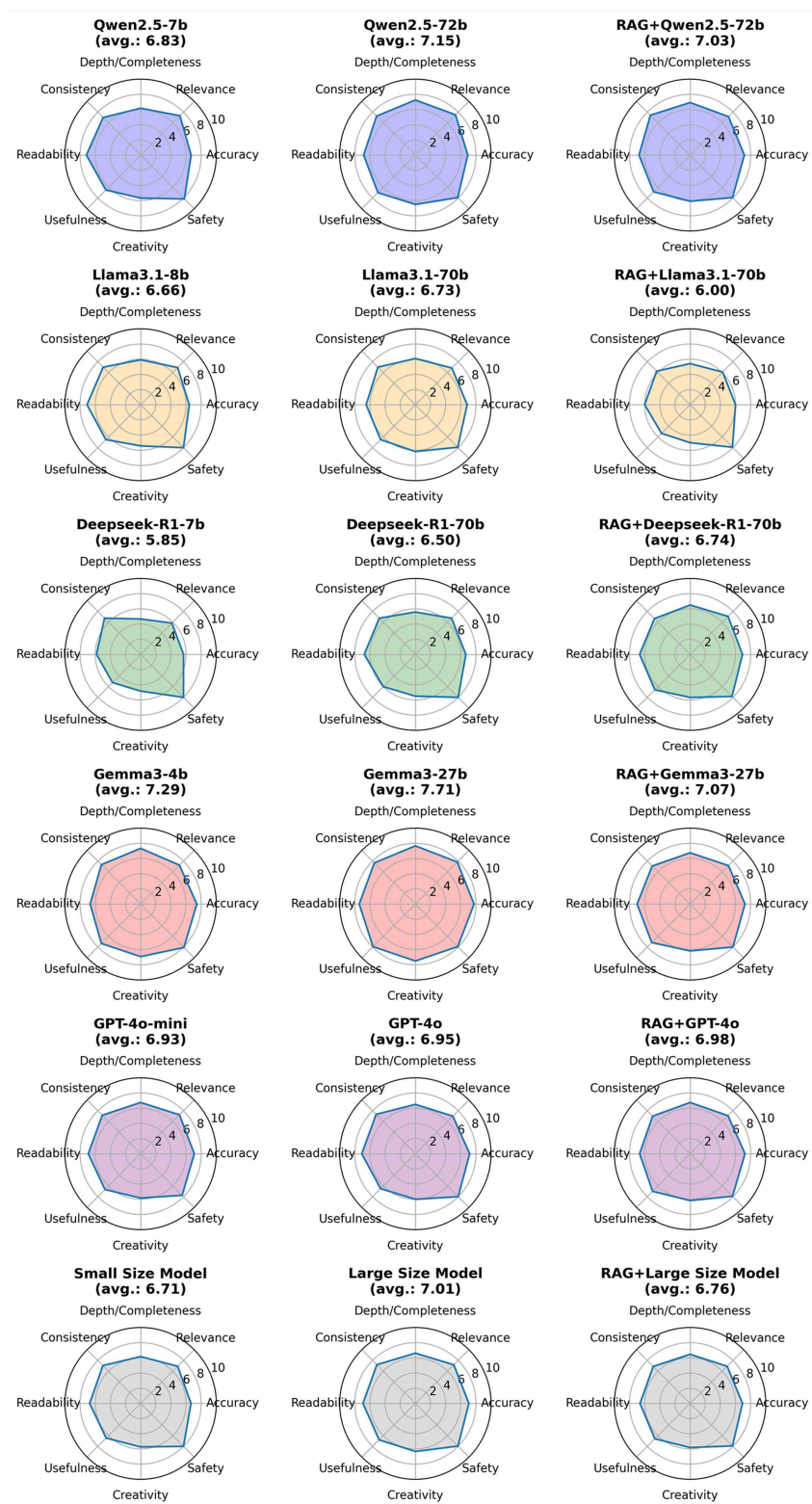


Figure 3

Human evaluation results of LLMs. Each radar chart demonstrates eight indicators for each model. The final results for all indicators are presented following the model's name.

Supplementary Files

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- [SupplementaryMaterial.docx](#)
- [SupplementaryTable1.xls](#)